Quantization of IQ Channel Data for Plane-Wave Fourier-Domain Beamforming

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***Abstract*—Fast Fourier-domain beamforming (FDB) is a highly efficient approach to plane-wave (PW) image reconstruction, but it needs to access each channel data frame all at once to take advantage of the Fast Fourier Transform (FFT) algorithm. To reduce the cost of probe-to-beamformer data transfers, we propose to compress complex-valued in-phase/quadrature (IQ) channel data using a simple quantization scheme that involves a small fixed codebook and a dynamic scaling factor updated at every snapshot of an array probe aperture during PW data acquisition. Our PICMUS-based results demonstrate that FDB can produce high-quality images after receiving only 4/4 bits per real/imaginary part of each IQ data sample. While this work is motivated by the FDB needs, our quantization scheme is also applicable in the case of delay-and-sum (DAS) beamforming.**

***Index Terms*—Plane-wave ultrasound imaging, IQ channel data, scalar quantization.**

I. Introduction

Plane-wave (PW) imaging facilitates ultrafast rates of channel data acquisition, which enables advanced biomedical diagnostics applications [1]. This modality is highly compatible with fast Fourier-domain beamforming (FDB) that performs image reconstruction in the frequency-wavenumber domain (e.g., see [2]), as opposed to the spatio-temporal domain used by conventional delay-and-sum (DAS) beamforming (e.g., see [3]). FDB methods, such as Temme-Müller (TM) migration [4]–[6], rely on full-frame data processing that imposes significant bandwidth and buffering requirements during probe-to-beamformer data transfers at high frame rates. Quantizing channel data samples prior to their transfer offers a practical solution to reduce this burden.

For complex-valued in-phase/quadrature (IQ) data, quantization can be applied to either the real/imaginary or magnitude/phase parts. We investigate both cases here, adopting the main ideas from [7] that dealt with scalar quantization of radio-frequency (RF) channel data. Unlike other, more advanced approaches with potentially higher compression ratios (e.g., see [8]–[12]), ours is extremely simple, as it aims at keeping compression-related computational resources and costs as low as possible. While motivated by the full-frame access requirements of FDB, the proposed method decompresses the quantized data back into IQ channel frames before reconstruction, thus ensuring general compatibility with other beamforming methods. Our approach has been evaluated using two PICMUS datasets [13] and considering two imaging cases: a single zero-angle PW and a 75-angle coherent PW compounding scheme (see Section III).

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II. Quantization Framework

Let $D_\theta^{\mathrm{IQ}}(t,x)$ denote a complex-valued 2D IQ channel-data frame, acquired by a linear transducer array after a PW emission steered at angle θ. Symbols t and x refer to temporal sampling instances and receive-element positions, respectively. The size of D_θ^{IQ} is given by $N_t \times N_x$, and each IQ sample can be expressed as follows:

$$D_\theta^{\mathrm{IQ}}(t,x) = R_\theta(t,x) + jI_\theta(t,x) = M_\theta(t,x)\cdot e^{jP_\theta(t,x)}, \quad (1)$$

where R_θ and I_θ are the real (in-phase) and imaginary (quadrature) IQ signal components, with M_θ and P_θ representing the corresponding magnitude and phase (between $-\pi$ and π).

Let N_1 and N_2 denote the original bit lengths of the two components (real/imaginary or magnitude/phase) of a given IQ data sample. During transfer, we want to use $M_1 < N_1$ and $M_2 < N_2$ bits instead. We rely on uniform scalar quantization, wherein two codebooks $\mathcal{C}_1$ and $\mathcal{C}_2$ are introduced:

$$\mathcal{C}_1 = \left(-1 + (k_1 + 1/2)\Delta_1 \mid k_1 = 0, 1, \ldots, 2^{M_1} - 1\right),$$

$$\mathcal{C}_2 = \left(-1 + (k_2 + 1/2)\Delta_2 \mid k_2 = 0, 1, \ldots, 2^{M_2} - 1\right),$$

where $\Delta_1 = 2^{1-M_1}$ and $\Delta_2 = 2^{1-M_2}$. Given an M_1-bit index k_1, we can get the corresponding entry $\mathcal{C}_1(k_1)$. Similarly, an M_2-bit index k_2 is needed to retrieve $\mathcal{C}_2(k_2)$. It is important to note that the codebook values in both $\mathcal{C}_1$ and $\mathcal{C}_2$ fall within the interval $[-1, 1]$.

In addition to these codebooks, we also utilize so-called pilots associated with individual snapshots of N_x samples captured across the aperture at every temporal sampling instance t [7]. In this work, we set each pilot $p_\theta(t)$ to the maximum magnitude value among all IQ data samples acquired by the transducer array at given time t:

$$p_\theta(t) = \max_x |D_\theta^{\mathrm{IQ}}(t,x)|. \quad (2)$$

Next, we describe how such pilots and codebooks are used to compress and decompress IQ channel data.

A. Real/Imaginary IQ Data Quantization

We assume that complex-valued IQ channel data are represented in the Cartesian format: $D_\theta^{\mathrm{IQ}} = R_\theta + jI_\theta$. At a given t, the probe acquires an N_x-element data vector forming the aperture snapshot. First, we normalize the real and imaginary components of these elements as follows:

$$R_\theta^*(x)|_t = \frac{R_\theta(t,x)}{p_\theta(t)}, \quad I_\theta^*(x)|_t = \frac{I_\theta(t,x)}{p_\theta(t)}, \tag{3}$$

Next, for each x, we determine index $k_1(x)$ such that $R_\theta^*(x) \in [-1,1]$ falls within the uniform partition subinterval represented by $\mathcal{C}_1(k_1(x))$. Similarly, we determine $k_2(x)$ for $I_\theta^*(x)$ to be used with $\mathcal{C}_2$. Then, all k_1 and k_2 indices, requiring $N_x(M_1+M_2)$ bits in total, along with the (N_1+N_2)-bit pilot $p_\theta(t)$ are transferred from the probe to the beamformer. The latter retrieves the corresponding codebook entries and computes the dequantized snapshot values as follows:

$$\widehat{R}_\theta(x)|_t = p_\theta(t) \cdot \mathcal{C}_1(k_1(x)) \approx R_\theta(t,x), \tag{4}$$

$$\widehat{I}_\theta(x)|_t = p_\theta(t) \cdot \mathcal{C}_2(k_2(x)) \approx I_\theta(t,x), \tag{5}$$

This transfer process is repeated for every snapshot, i.e., at every temporal sampling instance t. As the real and imaginary components typically have comparable numerical ranges and require the same bit allocations, we have $N_1 = N_2$, $M_1 = M_2$, and $\mathcal{C}_1 = \mathcal{C}_2$ (i.e., a single codebook is sufficient).

B. Magnitude/Phase IQ Data Quantization

We assume that acquired IQ channel data are converted to the polar format: $D_\theta^{\mathrm{IQ}} = M_\theta \cdot e^{jP_\theta}$. At a given t, the magnitude and phase components of the captured N_x-element data vector are normalized as follows:

$$M_\theta^*(x)|_t = \frac{M_\theta(t,x)}{p_\theta(t)}, \quad P_\theta^*(x)|_t = \frac{P_\theta(t,x)}{\pi}, \tag{6}$$

Next, for each x, we determine $k_1(x)$ and $k_2(x)$ for $M_\theta^*(x)$ and $P_\theta^*(x)$, respectively. Then, the probe sends all k_1 and k_2 indices along with $p_\theta(t)$ to the beamformer. Finally, the latter obtains

$$\widehat{M}_\theta(x)|_t = p_\theta(t) \cdot \mathcal{C}_1(k_1(x)) \approx M_\theta(t,x), \tag{7}$$

$$\widehat{P}_\theta(x)|_t = \pi \cdot \mathcal{C}_2(k_2(x)) \approx P_\theta(t,x), \tag{8}$$

This transfer process is repeated for every snapshot, i.e., at every temporal sampling instance t. The compression ratio is given by

$$\rho = \frac{N_x(N_1+N_2)}{N_x(M_1+M_2)+N_1+N_2}. \tag{9}$$

The above formula is applicable in the case of real/imaginary data quantization as well.

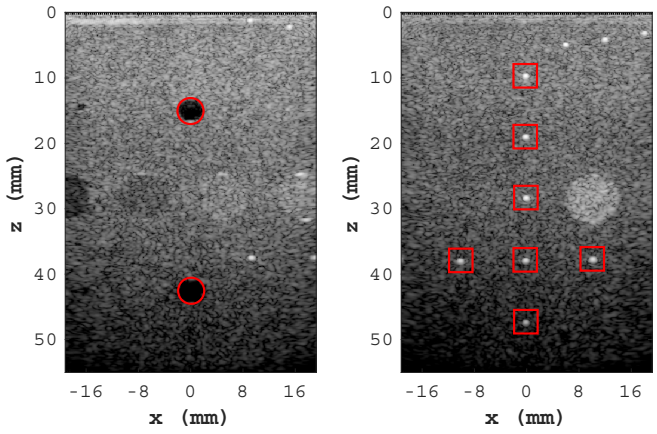


Fig. 1. DAS-beamformed TYPE-1 (left) and TYPE-2 (right) image examples with marked evaluation regions (75 plane waves, 60-dB range).

III. EVALUATION RESULTS

We evaluated our real/imaginary (RI) and magnitude/phase (MP) quantization schemes on two PICMUS *in-vitro* datasets [13], acquired by a 128-element probe and labeled TYPE-1 and TYPE-2 (see Fig. 1). For TYPE-1 images, we measured the generalized contrast-to-noise ratio (gCNR) values for both top and bottom marked anechoic targets. For TYPE-2 images, we measured the mean axial and lateral full width at half maximum (FWHM) values over seven marked point-like targets. These measurements are listed in Table I for two imaging cases: a single zero-angle PW emission, and 75 PW emissions at different steering angles $\theta \in [-16°, +16°]$. Both an IQ-based TM migration method [6] and a basic PICMUS-provided DAS beamformer [13] were assessed in terms of their reconstruction quality losses when using dequantized instead of original IQ channel data.

For the RI scheme, we considered three (M_1, M_2) settings: $(3,3)$, $(4,4)$, and $(5,5)$. Assuming the original data bit lengths of $N_1 = N_2 = 16$ and given $N_x = 128$ [13], the respective compression ratios ρ were equal to 5.12, 3.88, and 3.12. For the MP scheme, we set (M_1, M_2) to $(4,4)$, $(3,5)$, and $(2,6)$. Thus, the number of bits allocated to the phase component increased from 4 to 6 at the expense of using fewer bits for the magnitude part, while keeping the total bit budget fixed at 8 with $\rho = 3.88$.

In the single-PW case, our measured gCNR values improve consistently as the RI (M_1, M_2) bit allocation increases from $(3,3)$ to $(5,5)$. After TM migration, the original top-target gCNR baseline is at 0.878, dropping to 0.764 in the 6-bit RI $(3,3)$ case, whereas the 10-bit RI $(5,5)$ configuration performs better, achieving 0.862. Although the 8-bit RI $(4,4)$ setting produces a slightly lower top-target gCNR of 0.842 than the 10-bit configuration, it reduces the index payload by 2 bits per IQ data sample, yielding a highly favorable trade-off. For 8-bit MP settings (M_1, M_2), the $(3,5)$ allocation provides the highest (or tied for highest) gCNR for every evaluated target across both TM and DAS methods. For instance, the MP $(3,5)$ setting allows TM migration to achieve top-target gCNR of

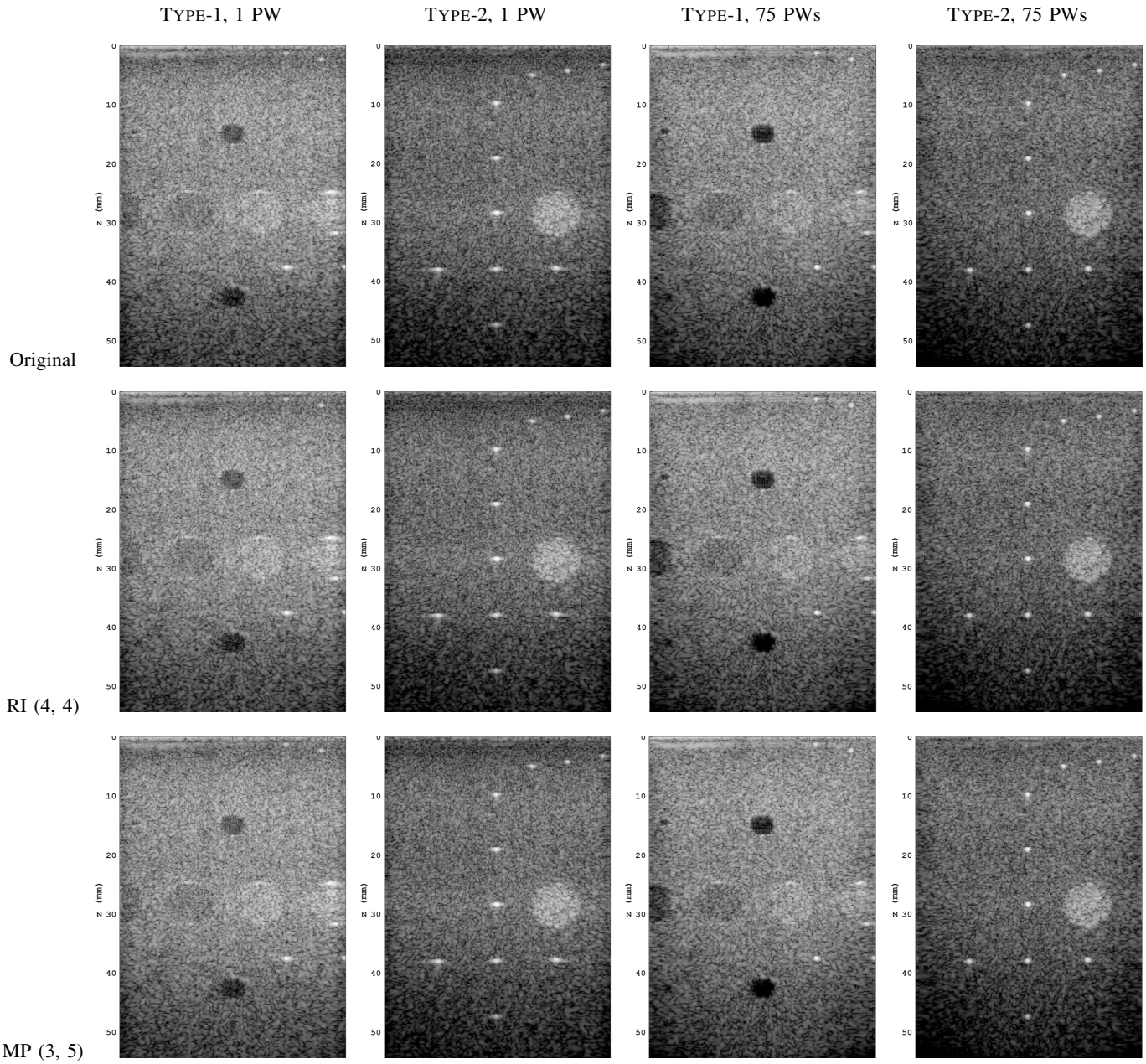


Fig. 2. TM-reconstructed images (60-dB range) using original and quantized IQ channel data. Row labels indicate data representation. Column labels indicate dataset type and PW count.

0.855, outperforming the other two cases, $(4, 4)$ and $(2, 6)$, that yield the gCNR values of 0.837 and 0.798, respectively. In other words, prioritizing the phase precision is beneficial; however, overly limiting the magnitude bit length results in contrast degradation. In terms of spatial resolution, measured FWHM values exhibit low sensitivity to quantization. The maximum absolute lateral FWHM deviation from the original baseline is approximately 0.011 mm. For TM migration, the baseline axial FWHM is at 0.482 mm, while all quantized variations fall between 0.482 mm and 0.490 mm.

In the 75-PW case, coherent compounding largely neutralizes contrast differences observed in the single-PW case. Every measured gCNR value stays within 0.007 of the uncompressed baseline, and every measured FWHM value remains within 0.005 mm. These results indicate that the RI and MP quantization schemes are well-suited for both Fourier-domain and conventional delay-and-sum beamforming.

TABLE I
IMAGE QUALITY ASSESSMENT FOR ORIGINAL, RI-QUANTIZED, AND MP-QUANTIZED IQ CHANNEL DATA PROCESSED BY TM MIGRATION AND DAS BEAMFORMING.

No. Waves	(M_1, M_2) bits	TM migration				DAS beamforming			
		gCNR_{top}	$\text{gCNR}_{\text{bottom}}$	$\text{FWHM}_{\text{axial}}$	$\text{FWHM}_{\text{lateral}}$	gCNR_{top}	$\text{gCNR}_{\text{bottom}}$	$\text{FWHM}_{\text{axial}}$	$\text{FWHM}_{\text{lateral}}$
1 PW	Original	0.878	0.831	0.482 mm	0.788 mm	0.849	0.815	0.482 mm	0.953 mm
	RI (3, 3)	0.764	0.707	0.483 mm	0.794 mm	0.771	0.760	0.485 mm	0.947 mm
	RI (4, 4)	0.842	0.809	0.485 mm	0.797 mm	0.822	0.811	0.485 mm	0.948 mm
	RI (5, 5)	0.862	0.827	0.482 mm	0.795 mm	0.827	0.813	0.485 mm	0.951 mm
	MP (4, 4)	0.837	0.773	0.485 mm	0.794 mm	0.809	0.780	0.483 mm	0.946 mm
	MP (3, 5)	0.855	0.804	0.485 mm	0.790 mm	0.827	0.780	0.485 mm	0.953 mm
	MP (2, 6)	0.798	0.775	0.490 mm	0.780 mm	0.818	0.766	0.487 mm	0.942 mm
75 PWs	Original	0.964	0.951	0.486 mm	0.526 mm	0.971	0.951	0.483 mm	0.613 mm
	RI (3, 3)	0.966	0.951	0.489 mm	0.522 mm	0.973	0.955	0.483 mm	0.611 mm
	RI (4, 4)	0.964	0.957	0.489 mm	0.525 mm	0.969	0.957	0.483 mm	0.611 mm
	RI (5, 5)	0.962	0.951	0.486 mm	0.525 mm	0.969	0.944	0.483 mm	0.611 mm
	MP (4, 4)	0.962	0.953	0.487 mm	0.528 mm	0.969	0.953	0.483 mm	0.611 mm
	MP (3, 5)	0.962	0.951	0.487 mm	0.525 mm	0.971	0.953	0.483 mm	0.611 mm
	MP (2, 6)	0.964	0.953	0.489 mm	0.526 mm	0.969	0.953	0.483 mm	0.611 mm

IV. CONCLUSION

This work demonstrated that snapshot-wise scalar quantization of complex-valued IQ channel data is an effective strategy for alleviating the data communication and storage bottleneck inherent to FDB techniques such as TM migration. By compressing data samples prior to probe-to-beamformer transfer and then decompressing them, this approach preserves complete compatibility with both Fourier-domain and conventional delay-and-sum beamforming. Our experimental comparison between real/imaginary and magnitude/phase quantization schemes revealed that allocating 8 bits (in total) per IQ data sample offers a good balance of compression savings and image fidelity, as illustrated in Fig. 2.

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